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Multimodal Prompt Injection: A Formal 4D Taxonomy for Image and Document Pipelines

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***Abstract*—Despite rapid advances in multimodal large language models (MLLMs), their cross-modal fusion architectures pose significant security concerns. Multimodal prompt injection (MPI) attacks enable adversaries to incorporate malicious instructions within non-textual inputs such as images, PDFs, and slides to manipulate model behavior, bypass controls, and exploit processes such as optical character recognition (OCR) and metadata parsing. Current research is fragmented due to a lack of a cohesive risk classification and consistent evaluation techniques. This paper introduces a formal 4D taxonomy $\mathcal{T} = \langle C, L, O, S \rangle$ that categorizes MPI attacks according to carrier type, pipeline location, attacker objective, and stealth level, resulting in 96 potentially unique attack classes and indicates that merely 7.3% are documented in existing research. Analysis indicates that no singular defense provides comprehensive coverage; a confirmed limit of 75% dimensional coverage is established across all deployable defenders. Layered architectures with provenance and standardized multimodal benchmarks are recognized as critical conditions for effective MPI mitigation.**

***Index Terms*—Multimodal Prompt Injection, LLM Security, Attack Taxonomy, Cross-Modal Threats, Adversarial Stealth**

I. Introduction

The shift of artificial intelligence from text-only systems to multimodal large language models (MLLMs) is one of the most significant leaps in recent AI history. MLLMs have enabled a wide range of high-value applications by integrating text, image, and structured document processing through cross-modal fusion architectures, which are systems that combine multiple types of data (modalities) together for analysis [15]. The global multimodal AI market was valued at USD 1.6 billion in 2024. It is expected to develop at a compound annual growth rate of more than 30% [17]. This reflects the rapid adoption across industries including banking, healthcare, legal services, and knowledge management. The cross-modal fusion process is central to this capability. In this process, modality-specific encoders are components that turn each type of input such as text or image into a numerical representation (embedding) in the same space for the core language model to use. This allows a unified understanding of text, images, and documents.

However, this architectural advancement introduces a substantially broader and largely unexplored attack surface. Multimodal prompt injection (MPI) attacks have emerged as critical: adversaries embed malicious instructions in non-textual inputs such as images, PDFs, or slides to hijack model behavior, bypass safety controls, and exploit pipelines like OCR, metadata parsing, and feature extraction [1], [11]. Unlike traditional text-based prompt injection, addressed by the security community, MPI attacks exploit the trust multimodal pipelines place in non-textual content and use channels that textual defenses cannot monitor. Prompt injections are ranked as the foremost risk by industry bodies such as OWASP [10], representing an even greater threat in multimodal contexts where payloads can evade human reviewers, but remain fully interpretable to preprocessing stacks [9].

This vulnerability is particularly acute in document-centric and agent-based deployments. In retrieval-augmented generation (RAG) systems, poisoned documents or images can induce persistent cross-turn manipulation without any user interaction [16]. Similarly, in agentic workflows, compromised tool outputs or retrieved content can propagate injected instructions across agents, resulting in cascading system compromise. Empirical evidence confirms that even widely deployed commercial MLLMs including GPT-4o and Claude 3 are susceptible to indirect injection strategies, with all tested models demonstrating significant behavioral manipulation across N=594 controlled attack scenarios including unauthorized actions and data exfiltration in controlled clinical evaluations [1].

Despite growing awareness of these risks, the existing literature remains fragmented and inconsistent. Most studies address discrete attack scenarios in isolation, employ incompatible evaluation methodologies, and lack a unified framework for classifying multimodal threat vectors [2], [3]. Unfortunately, no standardized multimodal security benchmark currently exists, and proposed defenses including input sanitization, context isolation, and inference-time alignment are modality-specific and do not generalize across the full attack surface. This fragmentation substantially impedes comparative analysis and systematic progress toward effective MPI mitigation.

The review concentrates on image and document-centric modalities, which constitute the primary attack surface in corporate and agentic deployments, and for which the security community presently lacks a cohesive classification system. Audio and visual modalities are postponed for further development. There are three major issues that can be addressed and to clarify these, we present a structured 4D taxonomy of

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multimodal prompt injection attacks, classifying threats across four interrelated dimensions carrier type, pipeline location, attacker objective, and stealth level, which provides a unified framework to facilitate systematic comparison, identify gaps, and guide future research in multimodal security.

II. Background and Preliminaries

This section delineates the architectural and formal foundations for the following taxonomy and defensive analysis. Specifically, it covers the modular MLLM design, the cross-modal fusion process, and the expansion of the attack surface from textual to multimodal inputs.

A. Multimodal LLM Architectures and Pipelines

Modern MLLMs employ a modular architecture that combines modality-specific encoders with a central language model backbone. Vision encoders, often Vision Transformer (ViT) or CLIP-based contrastive models, extract visual features from image inputs and project them into the embedding space of the language model. Layout-aware encoders, OCR systems, and specialized parsers handle document input while preserving structural information like tables, headings, and spatial relationships. These encoded representations are then combined with textual tokens using cross-attention or concatenation to enable unified cross-modal reasoning. Typical processing pipelines include four stages: modality-specific preprocessing, feature encoding, cross-modal fusion, and autoregressive generation. Popular implementations include proprietary systems such as GPT-4o and Gemini 1.5, as well as open-source models such as LLaVA, Qwen-VL, and CogVLM.

Despite its strengths, this architecture substantially expands the attack surface relative to text-only LLMs. Non-textual modalities introduce input channels that circumvent safeguards designed exclusively for textual content. Hidden instructions embedded in images or documents such as low-contrast overlays, steganographic encodings, or invisible PDF layers remain imperceptible to human reviewers yet are fully tokenized by preprocessing pipelines [1]. Furthermore, the semantic ambiguity inherent in cross-modal fusion frequently causes extracted non-textual content to be treated as factual description rather than potential instruction, obscuring the boundary between trusted and untrusted input [14]. In agentic and RAG deployments, this risk is further compounded: contaminated retrieved documents or tool output might disseminate injected instructions across agent interactions, facilitating persistent multi-step manipulation [13].

B. Attack Model and Threat Formalization

Let $\mathcal{M}$ represent a multimodal LLM functioning within an input space $\mathcal{X} = \mathcal{X}_{\text{text}} \cup \mathcal{X}_{\text{image}} \cup \mathcal{X}_{\text{doc}}$, where each modality is encoded and integrated into a common latent representation prior to generation. A prompt injection attack is characterized by the formulation of an adversarial input

$$X_{\text{attack}} = X_{\text{benign}} + \epsilon, \tag{1}$$

where $X_{\text{benign}} \in \mathcal{X}$ represents the legitimate input and ϵ is a malicious perturbation introduced into one or more modalities, resulting in

$$\mathcal{M}(X_{\text{attack}}) = Y_{\text{adversarial}} \neq Y_{\text{intended}} = \mathcal{M}(X_{\text{benign}}). \tag{2}$$

The perturbation ϵ is modality-specific: for image input, ϵ_{image} can manifest as adversarial pixel-level noise or steganographic payloads; for document input, ϵ_{doc} may comprise concealed text layers, metadata alterations, or layout-embedded directives. The covertness of an attack is quantified by a discernibility function.

$$P(\epsilon) \leq \delta, \tag{3}$$

where δ denotes a threshold for human detectability. Attacks with $P(\epsilon) \approx 0$ are imperceptible to human reviewers yet remain fully operative within the model pipeline. The adversary's objective is to induce adversarial behavior aligned with one of four categories: task hijacking, policy evasion, data exfiltration, or persistent manipulation, as formally defined in the taxonomy presented in Section IV. Figure 1 demonstrates

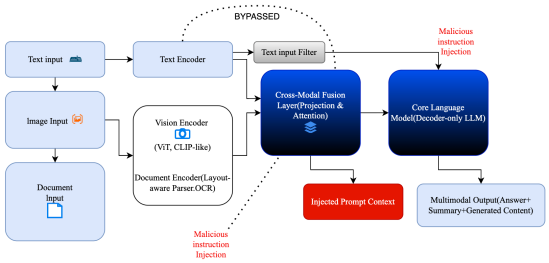


Fig. 1: Multimodal Prompt Injection Bypass Mechanism

how adversarial inputs manipulate the cross-modal fusion process to bypass conventional text-based defenses, leading to unauthorized model behavior or data exfiltration.

III. Literature Review

The integration of image and document processing capabilities to large language models (LLMs) improves multimodal reasoning for tasks such as optical character recognition (OCR), intelligent virtual assistants, and autonomous agents. However, this integration simultaneously increases vulnerabilities, particularly from multimodal prompt injection (MPI) attacks, which embed adversarial instructions within visual or document-based inputs. Therefore, recognizing these specific risks motivates researchers and practitioners to mitigate novel threats and develop defenses against evolving attack vectors.

Image-based attacks:Early MPI research focused on images. Clusmann et al. [1] demonstrated that embedded prompts in medical images successfully bypassed safety filters in GPT-4o and Claude 3 across N=594 controlled scenarios, resulting in unauthorized actions and data exfiltration with all tested models found to be susceptible regardless of overlay complexity. Pathade [11] introduced steganographic 'Invisible Injections,' that conceal adversarial instructions within images using spatial and frequency domain techniques, achieving

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24–32% success while remaining imperceptible ($PSNR > 38dB$). Li et al. [8] and Zhan et al. [17] investigated typographic attacks, demonstrating reliable hijacking of agent behavior through distorted typefaces or low-contrast text embedded in images. Ling et al. [9] further expanded image attacks to physical contexts, showing that real-world visual stimuli can prompt injections beyond digital environments, a dimension overlooked in prior work.

Document-based attacks: Similarly, another serious emerging threat is document-based injections. Yi et al. [16] demonstrated that OCR pipelines are vulnerable to metadata manipulation and obscured text in PDFs and slides, enabling indirect MPI in settings like scanned contracts and legal documents. Such attacks are particularly difficult to counter through post-processing, as the OCR pipeline fully tokenizes hidden instructions without leaving any visible trace. Zhan et al. [17] further benchmarked indirect injections within tool-integrated LLM agents, confirming that externally sourced documents constitute a primary risk in agentic workflows where models process unverified content.

Cross-modal attacks:Cross-modal attacks exploits both textual and visuals to increase impact. Wang et al. [13] showed that image manipulation plus text hijacking consistently alters multimodal agent decisions in RAG systems, with contaminated content enabling multi-turn persistence. Indirect MPI through documents or tool output spreads risks, as injected instructions pass among agents and interact [7]. Lee and Tiwari illustrated LLM-to-LLM prompt infections in multi-agent systems, showing that one compromised agent could transmit harmful directives system-wide via shared context.

TABLE I: Comparison of Related Work Against This Paper's Contributions

Ref.	Year	SLR	4D Taxonomy	Multi-modal	Formal Model
[3]	2026	✓	×	×	×
[2]	2026	✓	×	×	×
[14]	2026	×	×	×	×
[5]	2026	×	×	×	×
[15]	2025	✓	×	✓	×
[6]	2025	✓	×	×	×
[4]	2025	×	×	×	×
[12]	2025	×	×	✓	×
This work	**2026**	✓	✓	✓	- 4D taxonomy - Formal model - Image & doc. vectors - Pipeline threat modeling - Cross-modal propagation

Defenses remain limited and modality-specific. For example, clean token conversion using modified typography and advanced steganography can bypass input sanitization, including visual normalization and OCR filtering [1], [11]. Meanwhile, context isolation (sandwich prompting) helps separate trusted from external content but introduces significant usability drawbacks in content-dependent systems. Inference-time alignment and highlighting improve safety during generation but also introduce delays, increase computational complexity, and provide only limited protection. Furthermore, Syed et al. [12] introduced a provenance-based multimodal trust architecture combining modality-specific filtering and source tracking which is comprehensive but architecturally complex. In addition, Ferrag et al. [4] identified protocol-level vulnerabilities in agentic workflows that reactive protections do not address, exposing a gap in structural security for agentic deployment. Finally, layered techniques only cover some threats and do not generalize across the MPI attack surface.

Despite these advancements, the literature reveals significant inconsistencies. Many studies target isolated attack vectors image-only, typographic-only, or document-only threats without establishing a unified threat taxonomy [6]. Evaluation protocols are inconsistent; no standardized multimodal benchmarks mimic authentic document-centric environments or indirect RAG/agent operations. Systematic prompt injection analyzes predominantly address text-only or agent-level scenarios and lack a multimodal-specific threat taxonomy. The reviewed literature continuously overlooks four essential threat dimensions: payload carrier type, pipeline insertion location, attacker objective, and stealth level. Yeo and Choi [15] provide a related assessment of multimodal prompt injection threats and defenses, but rarely introduce a systematic dimensional taxonomy or formalize the underlying threat model. Defensive strategies remain reactive and modality-specific, frequently disregarding computational and usability constraints in enterprise contexts. The structured 4D taxonomy that is introduced in the following section is directly motivated by fragmentation gaps, which include undefined concealment levels, inconsistent threat vectors, and the absence of pipeline modeling.

IV. Proposed Taxonomy and Analysis

A. Taxonomy Design and Formal Definition

Current MPI research is disjointed. Studies often focus on individual attack vectors (specific ways attackers act), lacking a single taxonomic framework. Earlier taxonomies stress text-based injection (manipulating text input) [2], [6], [14] or broad agent-level threats (risks to the whole system or agent) [4], [5], However, they fail to address the specificity required for image and document-centric processes. To rectify this deficiency, we present a comprehensive 4D taxonomy that classifies MPI attacks across four interconnected dimensions: carrier type (how prompts are carried), pipeline location (where attacks enter data flow), attacker objective (describing the aim), and stealth level (how well detection is avoided).

The taxonomy is formally defined as a 4-tuple of sets:

$$\mathcal{T} = \langle C, L, O, S \rangle, \quad \mathcal{A} = (c_i, l_j, o_k, s_m) \in C \times L \times O \times S \quad (4)$$

Here,C is the set of carrier types, L is the set of pipeline locations, O signifies the set of attacker objectives, and S

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indicates the set of stealth levels. Each specific MPI attack is represented as a 4-tuple $\mathcal{A}$, with each component of a distinct dimension.

This Cartesian product framework serves three main functions. It enables comprehensive enumeration of attack categories, exact mapping of incidents onto the taxonomy, and identification of under-researched combinations. For example, Document + System Prompt + Persistent Manipulation + Invisible denotes a high-impact attack category lacking dedicated defenses in existing literature where $X_{\text{attack}} = X_{\text{benign}} + \epsilon$ and $\mathcal{M}(X_{\text{attack}}) = Y_{\text{adversarial}} \neq Y_{\text{intended}}$:

Carrier Type refers to how the perturbation ϵ is delivered or encoded. To hide data, image carriers can use methods like adversarial patches, steganographic embeddings, low-contrast overlays, and typeface alterations [9]. Document-based carriers ϵ_{doc} make use of hidden text layers, metadata modifications, and layout-integrated directives in formats including PDFs, presentations, and scans [16]. Different carrier types impact information encoding and detection.

Pipeline Location is where ϵ enters the multimodal processing pipeline. Sources include direct user upload (X_{upload}), external retrieval ($X_{\text{retrieval}}$), tool or API output (X_{tool}), and system prompt integration (X_{prompt}).Indirect paths, especially $X_{\text{retrieval}}$ and X_{tool}, present a higher risk. The model processes these without user knowledge.

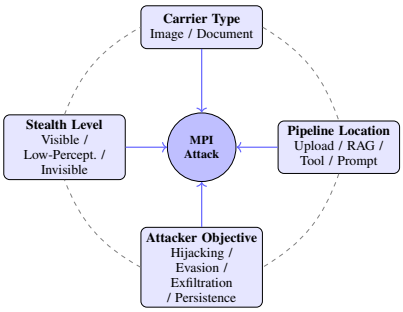


Fig. 2: The proposed 4D taxonomy framework for classifying multimodal prompt injection attacks. Each dimension characterizes a distinct property of the adversarial perturbation ϵ and its interaction with the multimodal pipeline.

Attacker Objective delineates the induced adversarial consequence, denoted $Y_{\text{adversarial}}$ which refers to the specific type of damage or outcome that results from an MPI attack. $Y_{\text{adversarial}}$ has four main types: goal hijacking (taking control of task output), policy evasion (using safety filters), data exfiltration (leaking sensitive data), and persistent manipulation (changing the behavior of the model for multiple users or between sessions) [7], [13].

Stealth Level Stealth Level is the level of easy detection of ϵ that is measured by $P(\epsilon) \leq \delta$.There are three levels. Visible ($P(\epsilon) > \delta$): changes are perceptible to human reviewers. Low-perceptibility ($P(\epsilon) \approx \delta$)($P(\epsilon) \approx 0$): changes use methods like steganography or hidden layers, with PSNR $>$ 38 dB and SSIM $>$ 0.94 [11]. Invisible attacks are the most severe threat; humans do not notice them, but the model does.

Figure 2 illustrates the four dimensions as an interconnected framework, showing how each dimension characterizes a distinct property of ϵ and its relationship to the others. Table II further complements Figure 1 by summarizing these dimensions, providing representative categories and example attack vectors for each.

TABLE II: 4D Taxonomy of Multimodal Attack Dimensions

Dimension	Categories	Example Vector
Carrier Type	Image, Document	Steganographic PNG, Layered PDF
Pipeline Location	Direct Upload, RAG, Tool/API, System Prompt	Poisoned web image, Tool output injection
Attacker Objective	Hijacking, Policy Evasion, Exfiltration, Persistent Manipulation	Redirecting output, Leaking system prompt
Stealth Level	Visible, Low-Perceptibility, Invisible	Overt text overlay, Distorted typography, Pixel-level steganography

B. Threat Models

The 4D taxonomy allows us to characterize three threat models. These models differ in privilege and sophistication, and each targets specific areas within the Cartesian product space of the Taxonomy $\mathcal{T} = \langle C, L, O, S \rangle$.

Black-box (End-User) attackers upload manipulated images or documents through X_{upload}, they target (c_i = Image, l_j = Upload, o_k = Hijack, s_m = Low-P) Clusmann et al. [1] demonstrated that all tested models including GPT-4o and Claude 3 were susceptible to embedded sub visual prompts across N= 594 controlled attack scenarios.

Gray-box (Content-Provider) attackers contaminate external sources using ($l_j \in \{X_{\text{retrieval}}, X_{\text{tool}}\}$, s_m = Invisible). This lets them bypass all user-facing filters. Yi et al. [16] showed near-complete hijacking using poisoned PDF retrieval. Wang et al. [13] found that poisoned knowledge base entries persist in interaction. No single-layer defense stopped these attacks during retrieval.

High-Privilege (Insider) attackers manipulate X_{prompt} and context assembly. They can activate the entire taxonomy: $\mathcal{A}$ = (Multi-modal, X_{prompt}, Persistent, Invisible) Lee and Tiwari [7] showed a compromise starting with one poisoned agent. Ferrag et al. [4] confirmed exploits at the protocol-level below all detection thresholds.

Mapping these attack levels into $\mathcal{T}$ shows a clear escalation pattern. Each higher privilege level adds more taxonomy dimensions. The black-box tier targets one location, while the insider tier spans the full Cartesian product. This means that no single defense can protect against all levels. For example,

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upload-path sanitization blocks black-box attacks, but gray-box retrieval injections bypass it. Inference-time alignment helps with insider threats but fails against stealthy attacks [12], [15].

C. Attack-to-Taxonomy Mapping

To identify comprehensive limitations in the current literature, we map each typical attack from Section III with the four dimensions of the taxonomy. This mapping reveals three essential insights. For example, carrier type is the most consistently examined dimension; nearly all studies specify whether their attack targets are image or document input. Conversely, researchers only partially explore pipeline location, with most focusing on direct upload scenarios. Higher-risk retrieval and tool output locations remain inadequately addressed. Concretely, of the 17 reviewed works, only 7 unique dimension combinations (c_i, l_j, o_k, s_m) were explicitly characterized, yielding a taxonomy coverage rate of $\frac{7}{96} \approx 7.3\%$ the first empirical fragmentation measure of its kind in the MPI literature.

The attacker's objective dimension, which defines the attacker's intent is consistently underdefined in the examined literature. Many studies implicitly assume goal hijacking, where an attacker changes the output to meet their goal. They do not expressly define policy evasion (bypassing access controls), data exfiltration (stealing private data), or persistent manipulation (long-term changes to system operation) as separate threat categories. These categories require distinct defenses. For example, a security measure designed to prevent hijacking, such as context separation (isolating user-generated content from system prompts), cannot guard against data exfiltration through indirect retrieval. Each type of attack uses distinct pipeline locations and varying levels of stealth.

Third, fewer than half of the evaluated studies address invisible stealth attacks, and this further highlights gaps in coverage across the taxonomy. These attacks form the highest-impact threat category, as validated at PSNR$>$ 38 dB and SSIM$>$ 0.94. Wang et al. [13] and Syed et al. [12] approach comprehensive dimensional coverage. However, neither offers the organized framework needed for systematic comparative analysis.

Finally, the most crucial under-explored area of $\mathcal{T}$ is the 4-tuple: Document + System Prompt + Persistent Manipulation + Invisible. This signifies a major attack category for which no tailored defenses exist. No benchmarks offer evaluative coverage. Importantly, this gap is visible only through the organized 4D framework and validates the taxonomy's analytical significance beyond simple classification.

V. Results and Discussion

A. Taxonomy Adoption Outcomes

The proposed 4D taxonomy $\mathcal{T} = \langle C, L, O, S \rangle$ provides businesses with a systematic threat modeling framework that enables clearer identification, classification, and prioritization of multimodal security concerns. A key benefit is its ability to generate 96 distinct attack classes via its Cartesian product structure, forming a comprehensive audit checklist. This allows organizations to quickly determine which attack classes are addressed by current defenses and which specific scenarios among the 96 remain unprotected, helping to focus security resources where they are most needed.

In document-centric and RAG-dependent workflows, the carrier type and pipeline location dimensions function as a formal risk assessment framework, exposing vulnerabilities in the$(\epsilon_{\text{doc}}, X_{\text{retrieval}})$ subspace that conventional defenses overlooked. Standard defenses typically fail to address vulnerabilities in this subspace. For organizations using multi-agent architectures, the taxonomy may show that the most significant attack surface that the (X_{tool}, X_{prompt}, Invisible) subspace remains neglected by their current security framework.

Utilizing the taxonomy as a standardized threat modeling framework facilitates a transition from reactive incident response to proactive assessment of dimensional coverage across the complete 96-combination space. It also helps to measure the improvement of security posture as new defenses are deployed.

B. Defense Adoption Outcomes

Using the dimensional coverage score as an analytical framework, each defense mechanism is systematically evaluated against all four taxonomy dimensions (C, L, O, S) yielding a structured score from 1/4 to 4/4 that reflects the number of dimensions a given defense effectively addresses. This framework serves as a practical metric for forecasting the potential real-world impact of defense investment choices without requiring empirical simulation.

Organizations relying solely on input sanitization or OCR post-processing are likely to reach only a 2/4 coverage ceiling. This may make stealth-level and indirect pipeline locations vulnerable. Enhancing inference-time alignment could raise the anticipated coverage ceiling to 75% and extend the coverage of the pipeline location. However, stealth-level threats likely remain, so steganographic and hidden-layer payloads continue to evade detection.

Achieving the 4/4 coverage threshold may require provenance-aware architectural strategies. These strategies would add trust labeling at the fusion layer, differentiating X_{upload}, $X_{\text{retrieval}}$, and X_{tool} before cross-modal encoding. A layered hybrid architecture including input sanitization, context isolation, and inference-time alignment—could reach a projected 3/4 coverage. This approach offers manageable overhead for enterprises concerned with architectural complexity.

C. Research Gaps and Future Directions

Four fundamental research priorities emerge from the taxonomy and defense adoption analysis. First, standardized multimodal benchmarks sampling systematically across all dimension combinations of $\mathcal{T}$ are critically absent current benchmarks cover an estimated 7.3% of the 96-combination space, precluding reproducible cross-study comparison and empirical coverage verification. Second, overcoming the 3/4 validated coverage ceiling requires targeted research into lightweight

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invisible-payload detection beyond PSNR and SSIM thresholds, and retrieval-path provenance verification before fusion-layer processing dimensions no existing defense adequately addresses. Third, principled hybrid composition frameworks grounded in $\mathcal{T}$'s Cartesian product structure are needed to formally guarantee 4/4 dimensional coverage, replacing current ad hoc defense combinations that provide no completeness assurance. Fourth, modality boundary enforcement within the cross-modal fusion layer designed to interrupt adversarial instruction propagation across agent context boundaries remains entirely unexplored and may constitute a structurally necessary prerequisite for secure multi-agent MLLM deployment at scale.

VI. Conclusion

This paper addresses three distinct aspects of multimodal prompt injection security through a comprehensive review and formal analysis. This work is fundamentally rooted in the 4D taxonomy $\mathcal{T} = \langle C, L, O, S \rangle$, which categorizes MPI attacks according to carrier type, pipeline location, attacker objective, and stealth level. This classification results in ninety-six theoretically distinct attack classes that double the representational capacity of prior agent-level taxonomies and expose the most critical undefended combination: Document + System Prompt + Persistent Manipulation + Invisible. This work further establishes that a significant number of theoretically possible attack classes remain either completely unstudied or partially explored, marking the first formal quantification of research fragmentation in the MPI field. Furthermore, the dimensional coverage score system demonstrates that no deployable defense currently in use achieves full dimensional coverage, with cross-modal agentic propagation being the highest risky class with no validated coverage among all defenses currently in use.

Advancing MPI mitigation towards a formally approved comprehensive standard represents one of the most significant unresolved issues in applied AI security. Four structurally necessary directions are needed to achieve this: principled hybrid composition frameworks that formally guarantee complete dimensional coverage; lightweight invisible-payload detection beyond current perceptibility thresholds; standardized multimodal benchmark sampling systematically across all ninety-six attack classes; and modality boundary enforcement within the cross-modal fusion layer to prevent adversarial instruction propagation across agent context boundaries.

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